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10/10/24

CIM612



**Icon Intuitiveness
Experiment**



Introduction

The purpose of this experiment was to conduct an evaluation of icon recognition, information scent, and icon attractiveness for icons found in the app for Apple Music. This experiment will inspect user mental models and provide useful data on opportunities for improving the icons in the future. In this experiment, we tested the intuitiveness of 11 different icons through a survey method that asked qualitative and quantitative subjective questions to participants relating to individual icons in a random order and then the icon set as a whole. This study found that some icons were better recognized than others and that some were deemed more attractive than others. Overall, I found three icon candidates to be redesigned due to their lack of recognition or their unattractiveness.



What is Apple Music?

Product Description

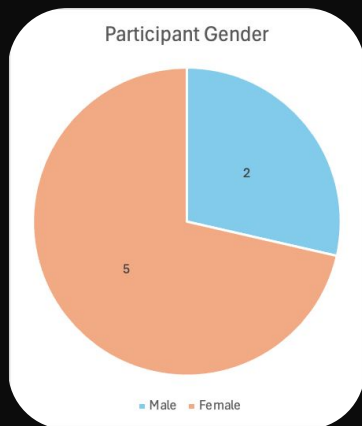
Apple Music is an audio and video streaming service developed by Apple Inc. Users select music to stream to their device on-demand, or they can listen to existing playlists. The service also includes the Internet radio stations Apple Music 1, Apple Music Hits, and Apple Music Country, which are broadcast live to over 200 countries 24 hours a day. The service was announced on June 8, 2015, and launched on June 30, 2015. New subscribers get a one-month free or six months free trial with the purchase of select products before the service requires a monthly subscription.



Participant Demographics

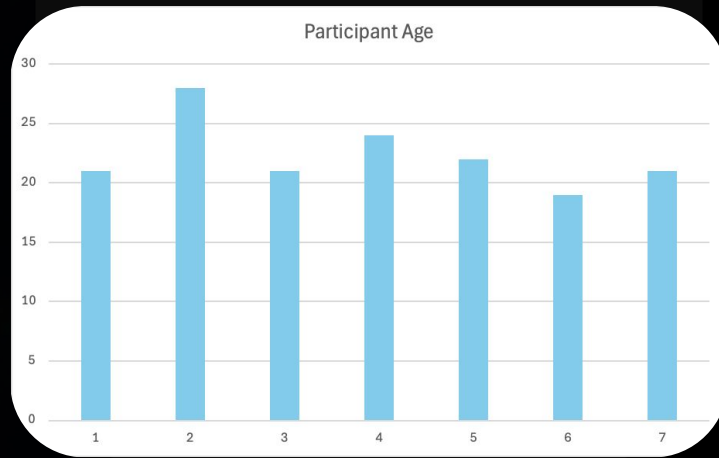
Gender

5 females and 2 males
with the majority of female
participants



Age

All participants were
between the ages of 17
and 30



Participant Experience

Education

Each participant was asked to list their education level which are listed below:

- Undergraduate: 4
- Graduate: 1
- Post-Graduate: 2

No one in the survey was in high school or in primary school.

Experience

- All participants had experience with apple products.
- 6 participants have used similar products while only 1 had not
- 5 participants were pursuing advanced studies relating to technology and 2 were consistent daily users of technology

The Icons Used in the Study



1



2



3



4



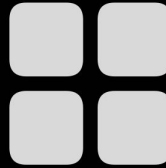
5



6



7



8



9



10



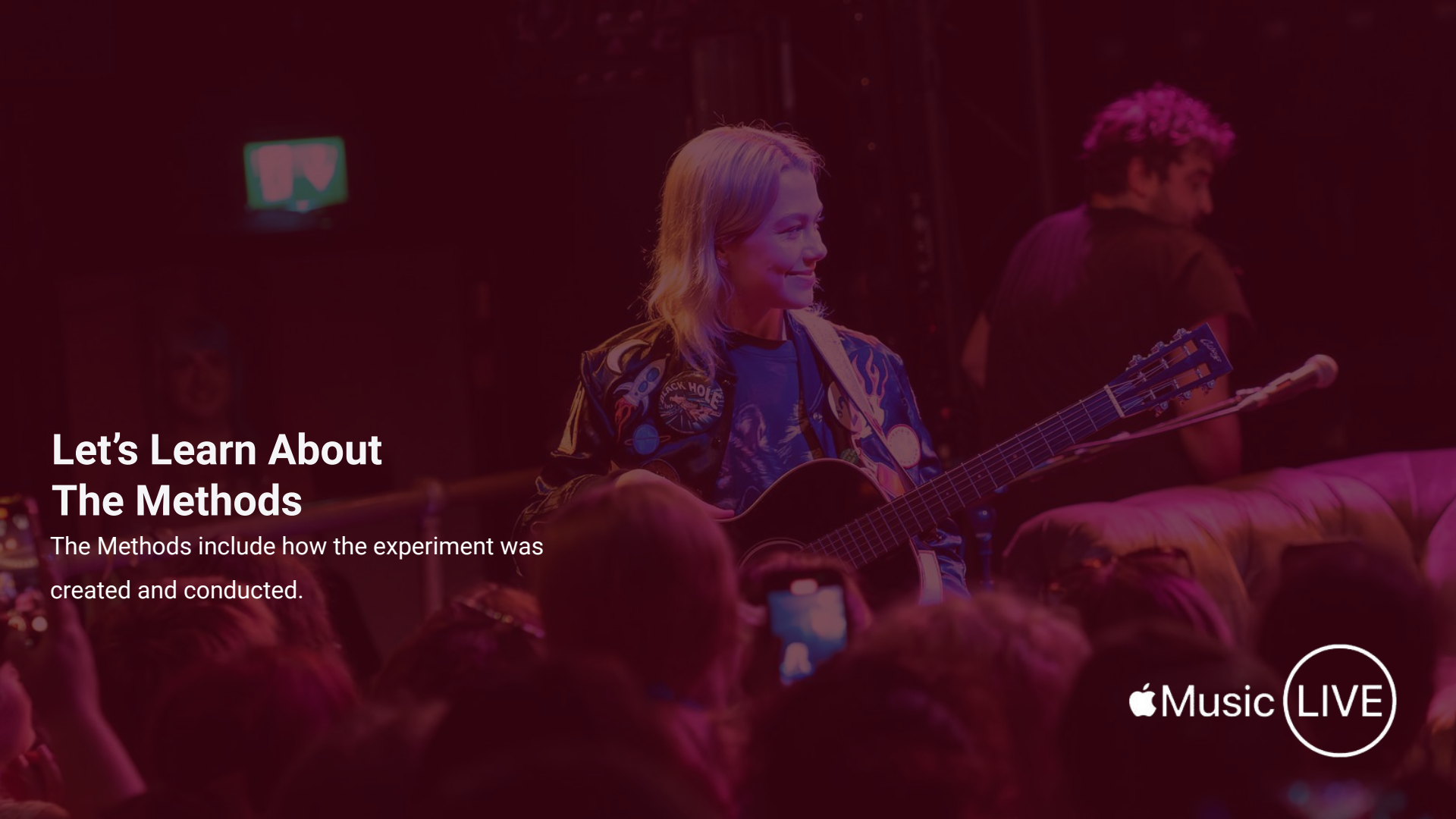
11

1. Play
2. Pause
3. Shuffle
4. Repeat
5. Autoplay
6. Lyrics
7. Home
8. New
9. Radio
10. Library
11. Search

Equipment Used

1. Slides
2. Qualtrics
3. Photoshop
4. SF Symbols
5. Excel



A woman with blonde hair is playing an acoustic guitar on a stage. She is wearing a blue t-shirt and a jacket with a colorful, abstract pattern. The background is dark with some stage equipment visible. The entire image has a red overlay. In the bottom right corner, there is a logo for Apple Music LIVE.

Let's Learn About The Methods

The Methods include how the experiment was
created and conducted.





Methods

Apparatus: Instructions & Pre-Questions

A survey was the best option for this study to gather quantitative and qualitative subjective data simultaneously. Using Qualtrics, a survey building platform, I was able to create a survey that first presented a page with the description of the survey, a description of the product, and an example questionnaire. After the first page, the participant went on to answer some pre-questions about demographic and prior experience with technology as explained to them on the first page. The second page would be essential in gathering information on the participants that would not be disclosed otherwise.

This experiment tests the intuitiveness of the icons used in an unnamed product. Participants will be given a brief description of the product followed by pre-questions, then questions about individual icons, and lastly, questions relating to the entire set of icons.

The product being tested in this experiment is a music streaming service that allows users to save songs to a library or to playlists. Users can access different genres, artists, and recommendations available in the app. Additionally, users can access radio shows, music videos, and lyrics that can all be saved offline across multiple devices.

Try to answer all questions to the best of your ability and there are no right or wrong answers!

Below I have written out a sample response:

The following questions are based on the icon below:



What would you describe this image as?

A: a fast forward button for music maybe.

What do you expect to happen when clicking on this icon in the product?

A: The current song will skip to the next song.

On a scale from 1 to 7, how attractive is this icon?

A: 3



How old are you?

Age

What is your gender?

☐ Male

☐ Female

☐ Prefer not to say

☐ Other

What is your experience with technology?

☐ Little to none

☐ Moderate weekly usage

☐ Consistent daily use

☐ Pursuing advanced studies relating to technology

Which education level are you?

☐ High School

☐ Undergraduate

☐ Graduate

☐ PhD or Higher

☐ Other

Have you used a product with this description before?

☐ Yes

☐ No

Which operating system are you comfortable with?

☐ Apple

☐ Android

☐ Both

☐ Other

Next page >



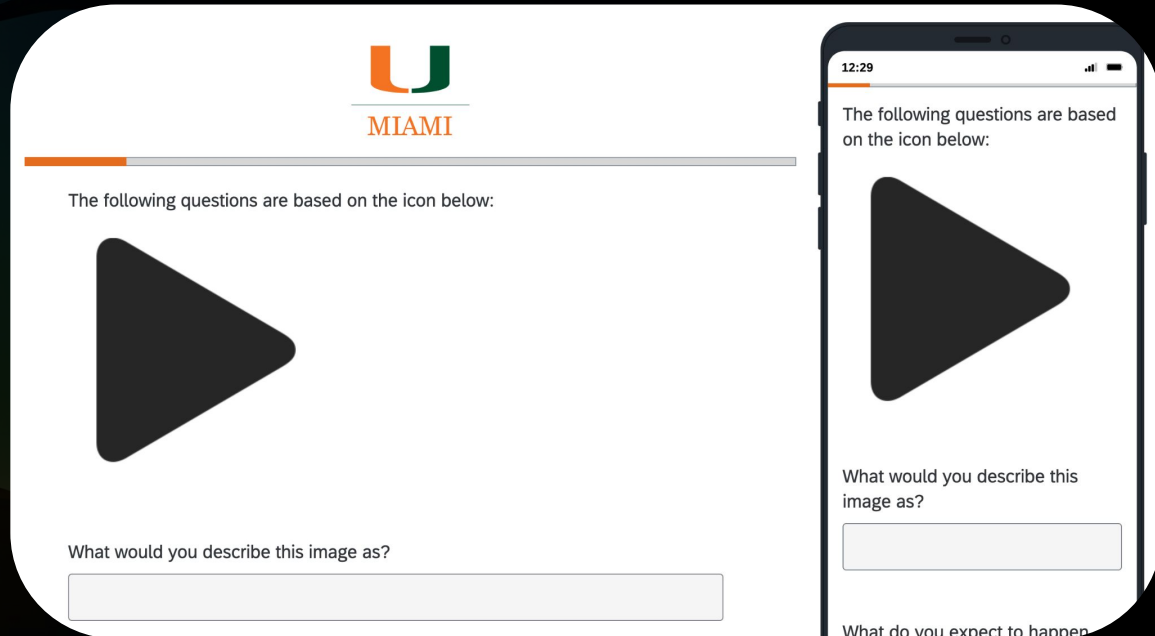
Methods

Apparatus: During Questions

For each icon, the participants were asked the following questions:

1. What would you describe this image as?
2. What do you expect to happen when clicking on this icon in the product?
3. Rate the attractiveness of this icon (1-7)

The icons were presented in random order and all their sizes were adjusted to have a diameter of 300 points or less vertically and horizontally so that they would be perceived equally. This bulk of the questionnaire provided the essential qualitative and quantitative data that would be used to evaluate the icons in relation to each other.





Methods

Apparatus: Post-Questions

The final set of questions were based on the icon set as a whole. All eleven icons were presented with a number to be easily identified. The participants were asked the following questions:

1. Which icon was MOST aesthetically pleasing (select 1-11)
2. Which icon was LEAST aesthetically pleasing (select 1-11)
3. Were there any aspects of the icons that you found confusing? If so, please specify which ones and why.
4. What would have made this/these icon(s) even better?

The survey ended with a thank you message.



The following questions are based on the image below:



Which icon was MOST aesthetically pleasing? (select 1-11)

MOST aesthetically pleasing

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 ☐ 11

12:29

The following questions are based on the image below:



Which icon was MOST aesthetically pleasing? (select 1-11)

MOST aesthetically pleasing

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 ☐ 11



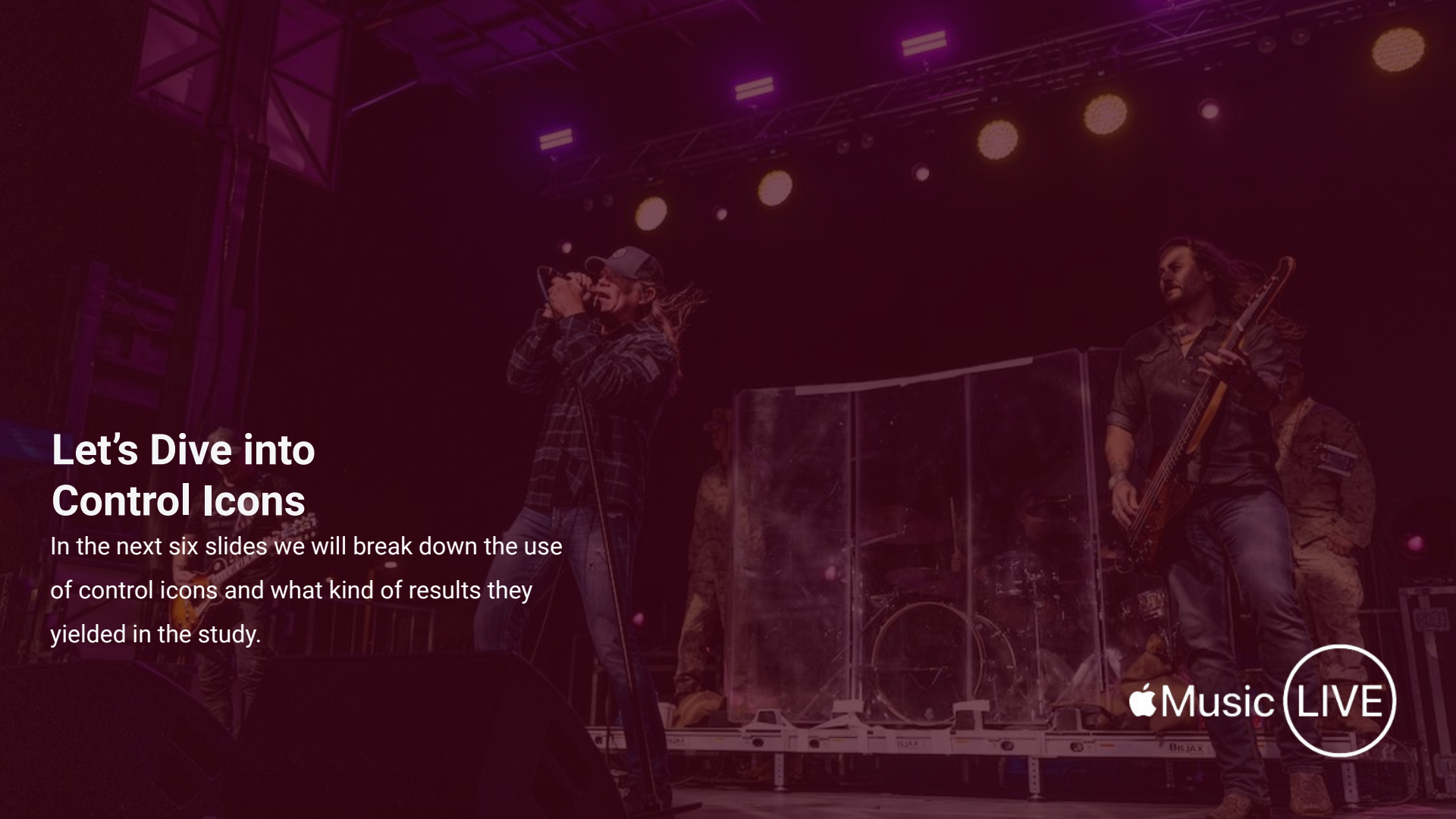
So What Were The Results?

The experiment yielded insightful results on the participants demographics, the icons' recognizability, and the icons' attractiveness



Summary of Icon Set

Icon	Description	Comments	Rating
	Play	"It will play something" " Plays a currently paused song, or starts a song."	6.1
	Pause	"the music/media playing will pause " "I expect the music to stop ."	6.1
	Shuffle	"I got confused between icons 3 and 4, as they are very similar" " Shuffles current playlist"	6.1
	Repeat	"I don't know the difference between 4 and 5" " Loop the music"	6.3
	Autoplay	"I couldn't tell what the icon was at all." "It was quite difficult to guess"	6.5
	Lyrics	" wasn't quite sure what the quotation stood for." "Shows song lyrics ?"	4.9
	Home	" Home page of library" "It will take users to the homepage "	6.3
	New	"I have no idea what this icon does" "I just really didn't know what the image represented"	3.7
	Radio	"This icon gives me anxiety " " Radio option will show up"	5.7
	Library	" Library will pop up" "It will take users to their music catalog "	5.9
	Search	"I will be able to search for specific music" " Search bar appears"	6.3



Let's Dive into Control Icons

In the next six slides we will break down the use of control icons and what kind of results they yielded in the study.

Apple Music **LIVE**

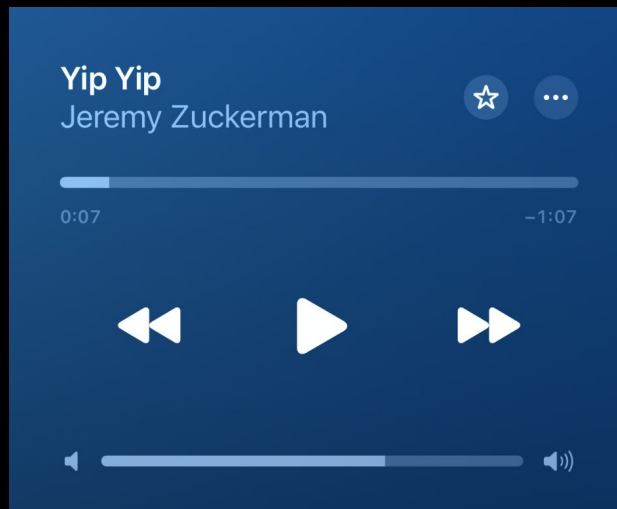
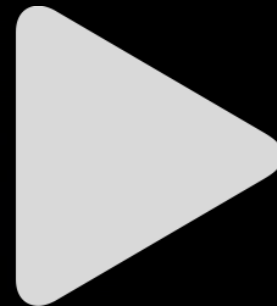
1. Play

7/7 correctly identified the icon

7/7 correctly defined the icon

Average rating: 6.1

Comments: None



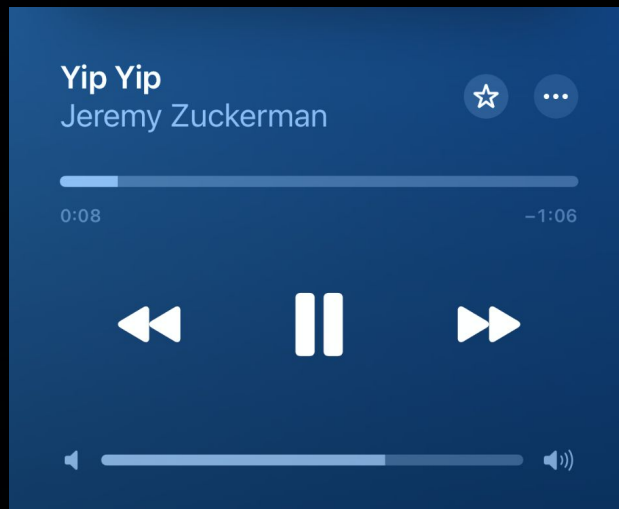
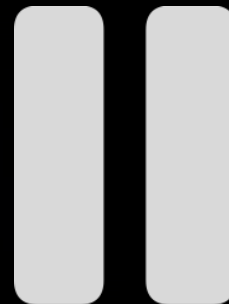
2. Pause

7/7 correctly identified the icon

7/7 correctly defined the icon

Average rating: 6.1

Comments: None



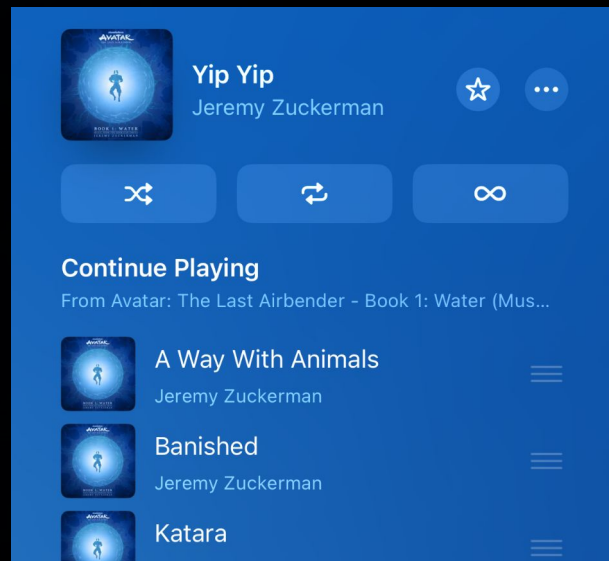
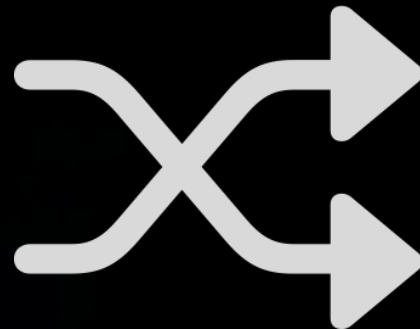
3. Shuffle

7/7 correctly identified the icon
7/7 correctly defined the icon

Average rating: 6.1

Comments:

"I got confused between icons 3 and 4, as they are very similar"



4. Repeat

5/7 correctly identified the icon

5/7 correctly defined the icon

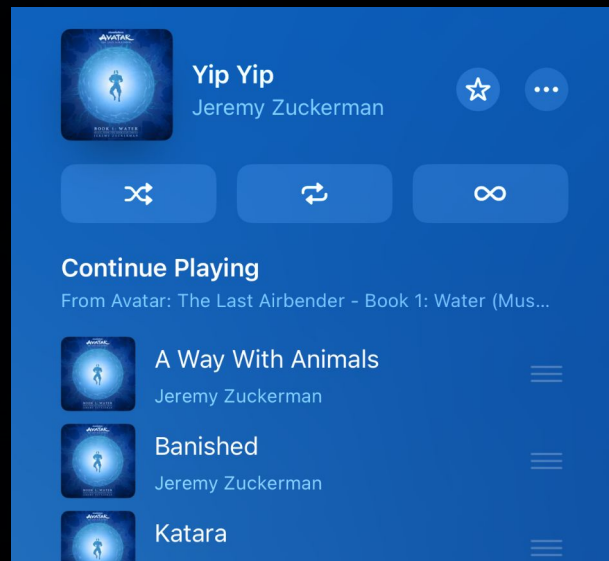
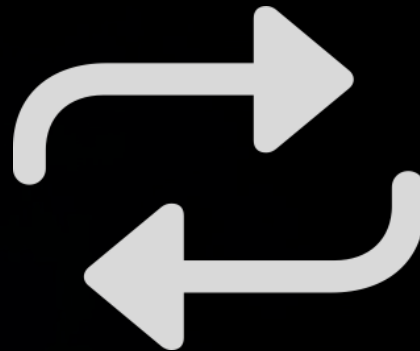
Average rating: 6.3

Comments:

"I am expected to refresh the page that I opened."

"I got confused between icons 3 and 4, as they are very similar"

"I dont know the difference between 4 and 5. They look similar."



5. Autoplay

1/7 correctly identified the icon

2/7 correctly defined the icon

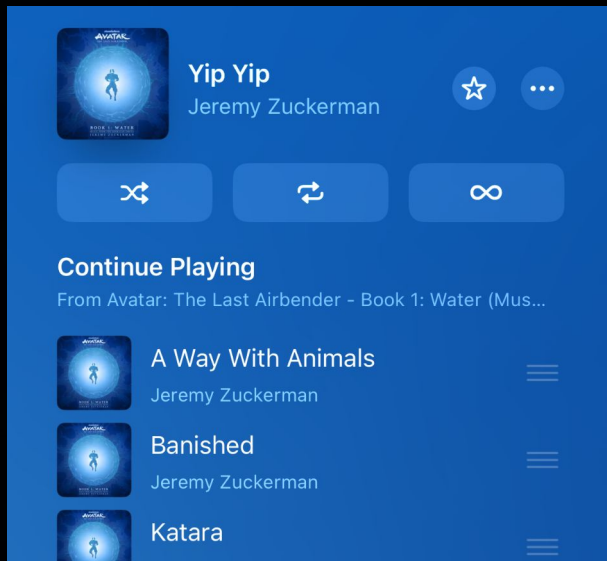
Average rating: 6.5 (received highest rating)

Comments:

“I am expected to share my item with someone”

“It is used to connect something (not sure what, but maybe connect with a device or other people)”

“Icon 5 confused me because I couldn't tell what the icon was at all.”





Baby Shark
Pinkfong



doo, d-doo

Baby shark, d-doo, d-
doo, d-doo

Baby shark

Baby shark, d-doo



0:23

-1:14



6. Lyrics

3/7 correctly identified the icon
3/7 correctly defined the icon

Average rating: 4.9

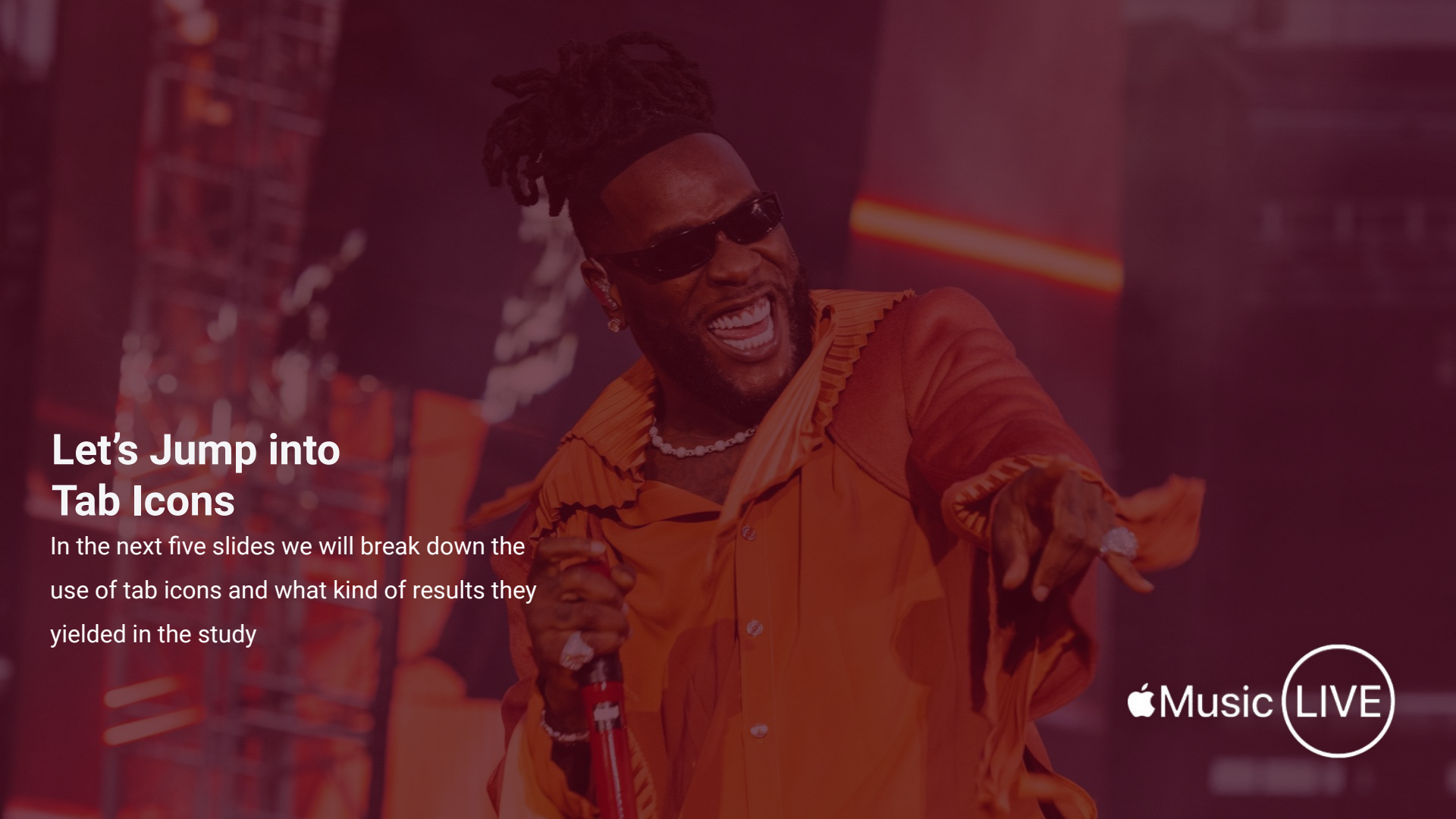
Comments:

"I am expect to receive some message"

"Maybe you can ask questions and get answers in this chat with the community that uses the product"

Icon 6 was a little confusing because I wasn't quite sure what the quotation stood for."



A man with dreadlocks and sunglasses, wearing an orange shirt, is performing on stage. He is holding a microphone in his right hand and pointing with his left hand. The background is a blurred cityscape at night.

Let's Jump into Tab Icons

In the next five slides we will break down the use of tab icons and what kind of results they yielded in the study



7. Home

7/7 correctly identified the icon

7/7 correctly defined the icon

Average rating: 6.3

Comments: None



8. New

4/7 correctly identified the icon

4/7 correctly defined the icon

Average rating: 3.7 (received the lowest rating)

Comments:

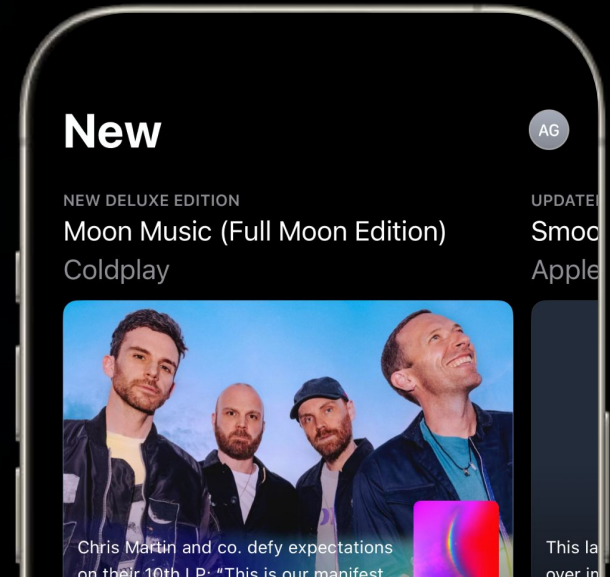
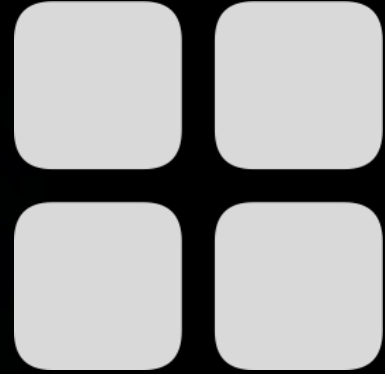
"I would expect to see other applications"

"I'm not sure"

"I have no idea what 8 does"

"Icon 8 was confusing because I just really didn't know what the image represented."

"I had no clue able number 8"



9. Radio

4/7 correctly identified the icon

4/7 correctly defined the icon

Average rating: 5.7

Comments:

“maybe you can change where you are listening to your music from, a speaker/earbuds/phone speaker system”

“Shows podcasts”

“number 9 gives me anxiety”

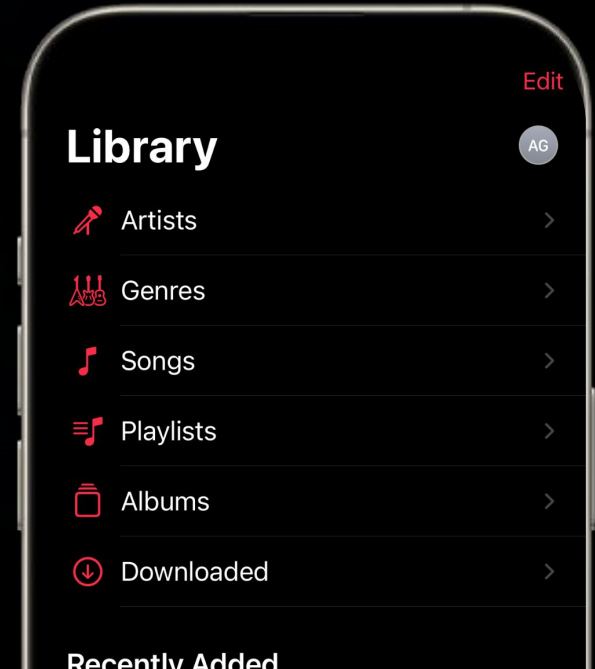


10. Library

6/7 correctly identified the icon
7/7 correctly defined the icon

Average rating: 5.9

Comments:
“could be Itunes? store”



11. Search

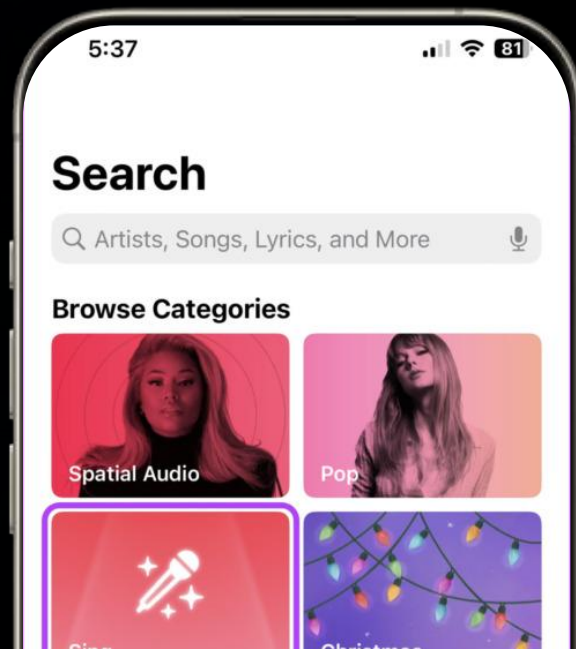
6/7 correctly identified the icon

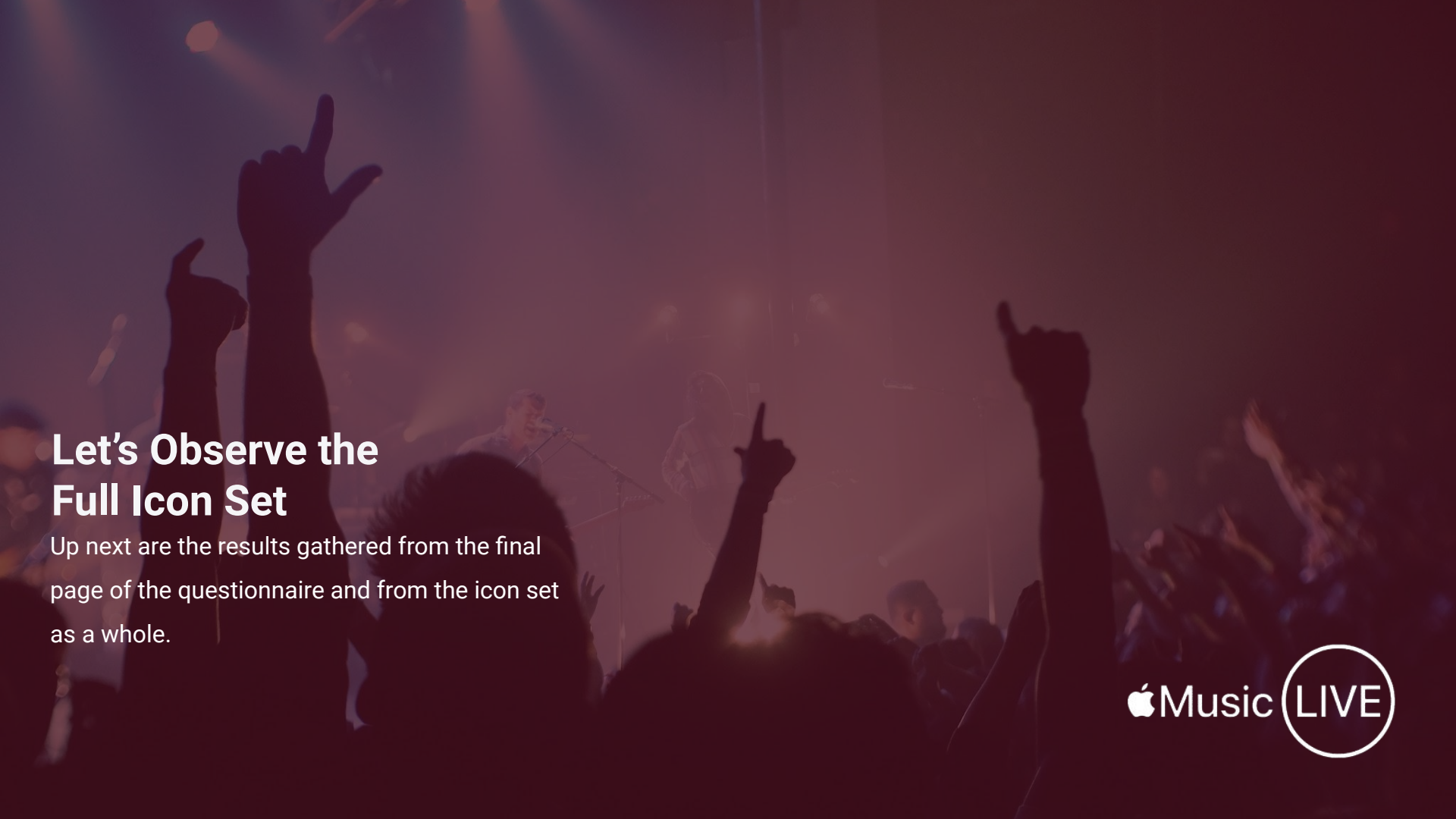
6/7 correctly defined the icon

Average rating: 6.3

Comments:

"I am supposed to enlarge my page"



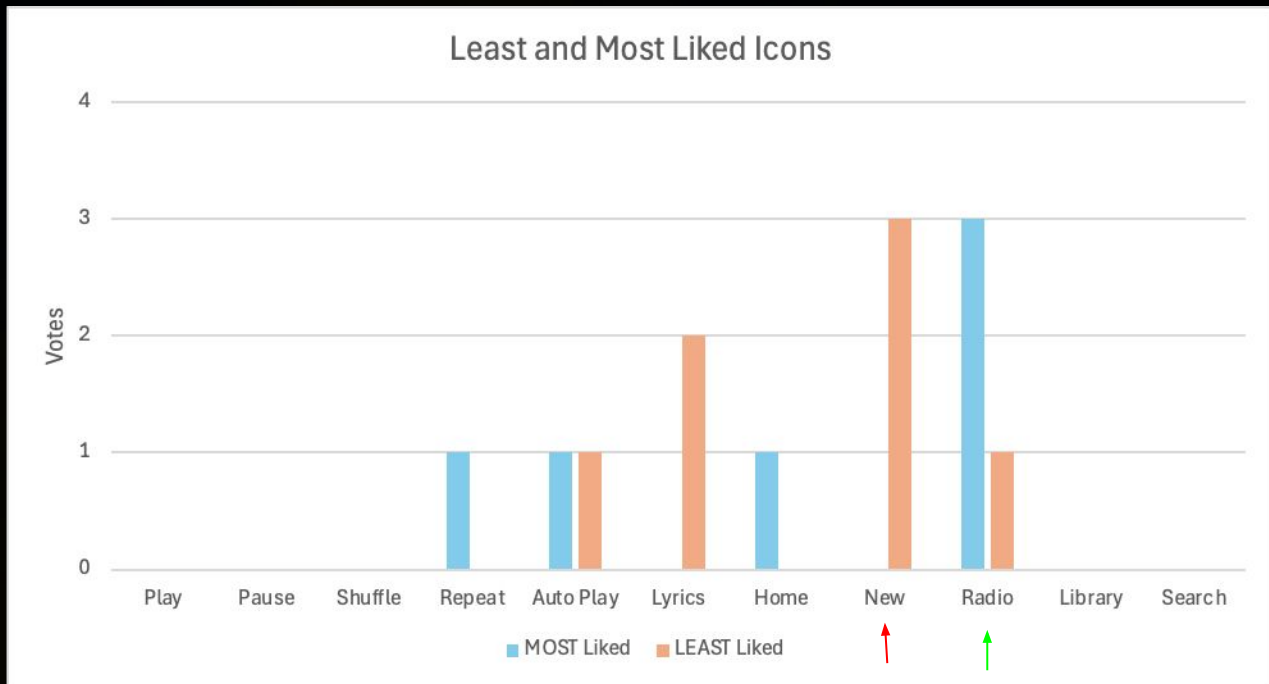


Let's Observe the Full Icon Set

Up next are the results gathered from the final page of the questionnaire and from the icon set as a whole.



Overall Least and Most Liked Ratings



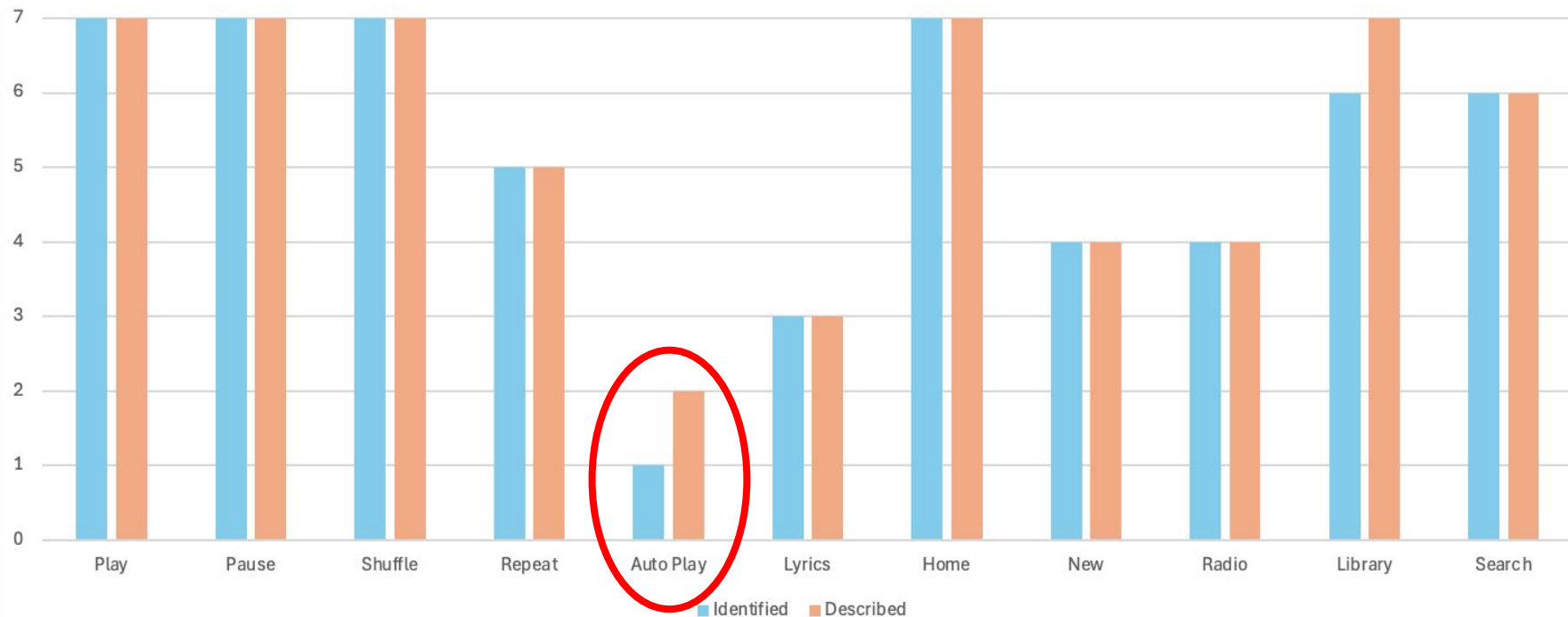
While some icons received no votes for least and most liked, icons like Autoplay and radio received votes for both liked and disliked. But the ones voted highest were New for least liked and Radio for most liked.

Overall Attractiveness Ratings



Overall Identified and Described Icons

Correctly Identified and Described Icons



Redesigning the Chosen Icons

With feedback and context, we can redesign the chosen icons to better reflect mental models of the users.



Icons Up for Improvement

Auto Play



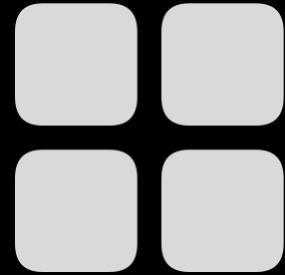
Although autoplay performed well with attractiveness, it failed in almost every trial in recognition.

Lyrics



With a high attractiveness score and low recognizability, lyrics can be improved upon.

New



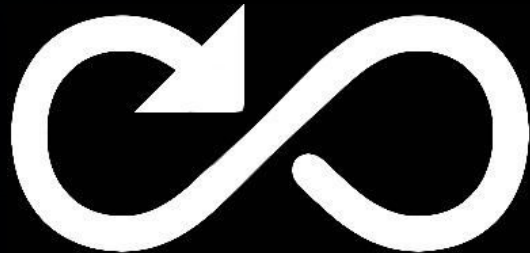
The new icon had issues with recognition and was rated the lowest in attractiveness.

Autoplay Redesign

The autoplay button automatically plays algorithmically suggested songs after the current playlist or album has ended.

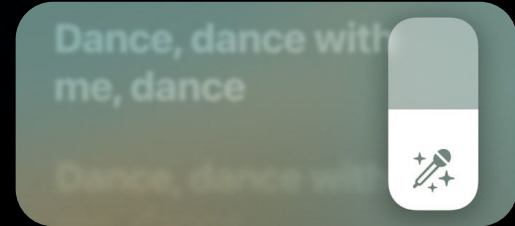
In context, the autoplay button is near the shuffle and repeat buttons. Incorporating the arrow element of this design can insinuate how it is similar to the other buttons but has an infinite element to it.

The final design is to the right.



6. Lyrics

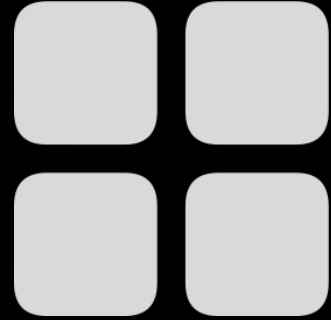
Many comments suggested that the quotation marks were confusing. Luckily, Apple has another feature similar to lyrics called “sing” which allows users to mute the vocals in a song and follow along with lyrics similar to karaoke. The icon for this feature is a microphone with stars around it. So the new icon for lyrics can be a microphone without any stars around it indicating that the full features of “sing” are not enabled.



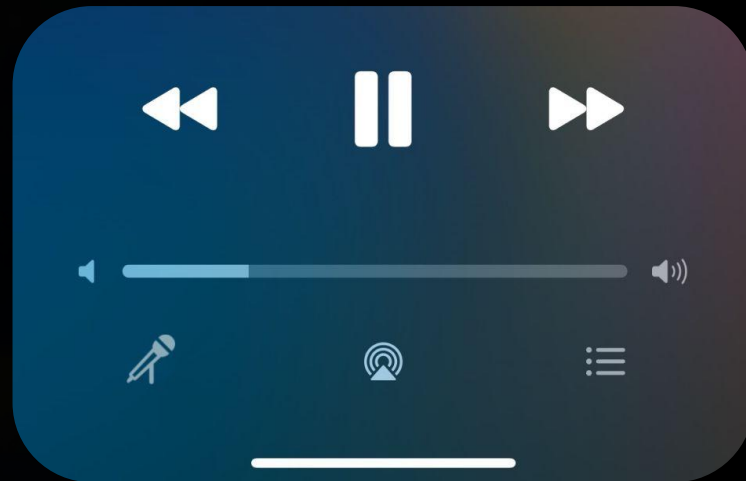
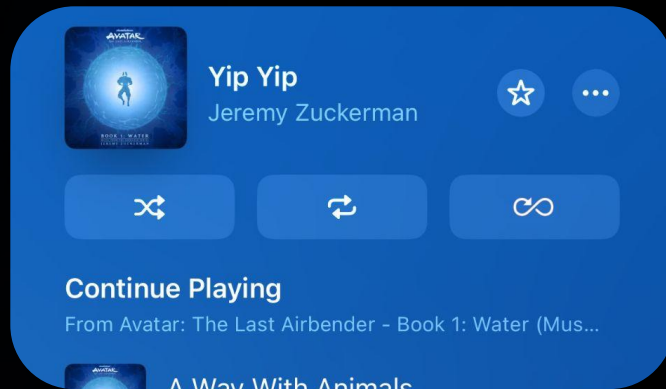
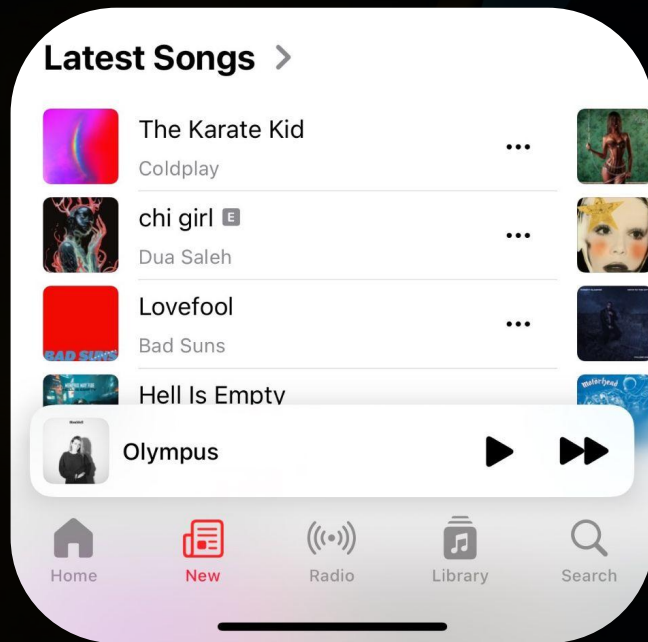
New

The new feature is a tab that displays new music, playlists, and other releases for the users to stay up to date with the current trends in music genres of their liking.

The main complaint about this icon is that it was unclear and very generic. The users cannot understand the icon's meaning. A new icon can look more like the front page of a newspaper which would match a more commonly understood mental model rather than just four squares.



In Context Redesigns



Discussion

Overall, this experiment showed how attractiveness and recognizability are not directly correlated and redesigns need adjust according to the principles that the icons fail to depict.

- **Redesign 1 Autoplay:** The new icon resembles the old icon because of its high rating in attractiveness and the added arrow helps to better fit the mental model that this symbol is related to the repeat and shuffle icons with an infinite element.
- **Redesign 2 Lyrics:** This icon pulls from the “sing” feature because it is so closely related to the lyrics feature
- **Redesign 3 New:** The new icon resembles the front of a newspaper which could better align with user mental models of how newspapers bring fresh and new information on a consistent basis

Conclusion

Throughout this experiment, we have found the relative attractiveness and recognizability of icons out of context through a survey method. This method incorporated quantitative and qualitative analysis that allowed the experimenters to analyze data to find the icons of least importance of the set. What we learned is that users have mental models developed from everyday activities that can be used to better design icons in a more intuitive fashion. Additionally, just because an icon is attractive does not mean that the icon is also recognizable and designs should encapsulate both concepts. This experiment raises more questions about context and interactivity that cannot be concluded from this experiment alone. Future research can add independent variables to the survey and have a larger set of participants.

Future Work

There are many variables at play in this experiment and some can be converted into independent or manipulated variables.

- Icon in context
- Icon color
- Device used
- Mobile, Desktop, or CarPlay

All of these variables can be tested in future experiments to understand whether the icons are recognized and liked to the same degrees when experienced closer to how they would be experienced in real life. Overall, these variables can only become confounding variables in the current experiment and allow the future of this research to grow.



Resources

[Apple Developer Icons](#)

[Apple Music Developer Icons](#)

[Nielsen Icon Usability Tests](#)

[Slidechef Apple Music Template](#)

